

URSHASHI MAJUMDER

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[Github](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Final-year B.Tech CSE (AI) student building Generative AI (GenAI) applications and backend software, with experience in agentic RAG, multi-agent workflows, Large Language Model (LLM) evaluation, Node.js/TypeScript microservices, REST APIs, databases, and React. Experienced with Docker and CI/CD for end-to-end development and deployment. Seeking software engineering, AI/GenAI, and full-stack roles.

EDUCATION

B.Tech in Computer Science Engineering (Artificial Intelligence), Techno India University 2023 – 2027
CGPA: 8.7 | Coursework: OOP, DSA, DBMS, Operating Systems, Computer Networks, Software Engineering

EXPERIENCE

Machine Learning & Artificial Intelligence Virtual Intern, IBM Aug 2025 – Jan 2026

- Built a salary prediction system using Linear Regression and Random Forest; evaluated performance using R^2 and MAE.
- Preprocessed data and trained models using Python, Pandas, NumPy, Scikit-learn, and Matplotlib.

PROJECTS

HR Policy RAG Assistant | Agentic RAG with Guardrails | [Github](#)

Python, LangChain, Qdrant, Jina Embeddings, Portkey, LangSmith, OpenEvals, Docker, GitHub Actions, AWS

- Built an agentic RAG assistant using LangChain tool calling, Jina embeddings, and Qdrant Cloud with top-3 retrieval to answer HR queries grounded in policy documents.
- Implemented input/output guardrails with gpt-oss-safeguard-20b to detect prompt injection, PII leakage, and unauthorized commitments; routed LLM calls through Portkey with fallback, retries, and caching.
- Evaluated correctness and groundedness using OpenEvals and LangSmith LLM-as-a-judge on a 10-question dataset; added tracing, structured logging, and Docker/GitHub Actions CI/CD for AWS ECR and EC2 deployment.

AI Research Agent | Multi-Agent Research Pipeline | [Github](#)

Python, LangChain, Mistral AI, Tavily, BeautifulSoup, Streamlit

- Built a multi-agent research pipeline with Search, Reader, Writer, Critic, and Revision stages, using Tavily for web search to generate structured, source-cited reports.
- Engineered role-specific prompts for each agent and implemented LLM self-critique with human-in-the-loop revision through a Streamlit interface, with Markdown report export.

Microservice Blog App | Event-Driven Full-Stack Platform | [Github](#)

Next.js, Node.js, Express.js, TypeScript, PostgreSQL, MongoDB, Redis, RabbitMQ, Gemini API, Docker, Tailwind CSS

- Built three independent Node.js/TypeScript microservices for user, author, and blog functionality with JWT authentication and PostgreSQL/MongoDB persistence.
- Implemented RabbitMQ-based event-driven cache invalidation with Redis cache-aside reads and integrated three Gemini-powered writing features.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, Java, SQL

AI / GenAI: Gen AI, Large Language Models (LLMs), Agentic AI, RAG, Prompt Engineering, Multi-Agent Systems, Tool Calling, Embeddings, Vector Search, Guardrails, LLM Evaluation

AI Frameworks & Tools: LangChain, LangGraph, LangSmith, OpenEvals, Qdrant, Portkey, Jina Embeddings, Tavily, Streamlit, Gemini, Groq, Mistral

Backend & Databases: Node.js, Express.js, REST APIs, Microservices, Flask, PostgreSQL, MongoDB, MySQL, Redis, RabbitMQ, JWT, Sequelize, Mongoose

Frontend: React, Next.js, HTML, CSS, Tailwind CSS

ML & Data: Scikit-learn, Pandas, NumPy, Matplotlib

DevOps & Tools: Git, GitHub Actions, Docker, Docker Compose, AWS (ECR, EC2), Linux, Postman, Swagger

ACHIEVEMENTS

- Finalist, Smart India Hackathon 2023
- 2nd Place, Plot2Deploy Hackathon 2024 (Google Developer Student Clubs)
- 5th Place, Hack Horizon 2.0, Arka Jain University (2026)